

MEDIATEK

MT8676 Hypervisor Clock User Manual

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Version History

Version	Date	Author	Description
1.0	2024-10-28	Guanghai Wang	Official release
1.1	2024-11-12	Guanghai Wang	Added Tbox in Section 1.2.2 SW Architecture

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1 Clock Common Interface

1.1 Overview

This chapter mainly introduces the MT8676 Clock architecture.

1.1.1 Abbreviations

Table 1-1. Abbreviations

Abbreviation	Explanation
CCF	(Linux) Common Clock Framework
CG	Clock Gate
Divider	Divider
MUX	Multiplexer
PLL	Phase-Locked Loop

1.2 Architecture/Process Overview

1.2.1 HW Architecture

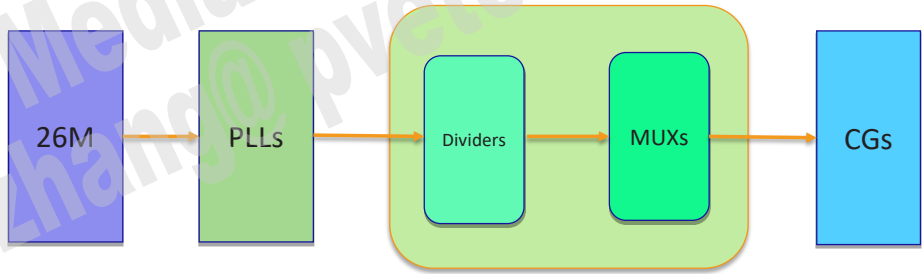


Figure 1-1. HW architecture

1.2.2 SW Architecture

1.2.2.1 Yocto/Android/Tbox

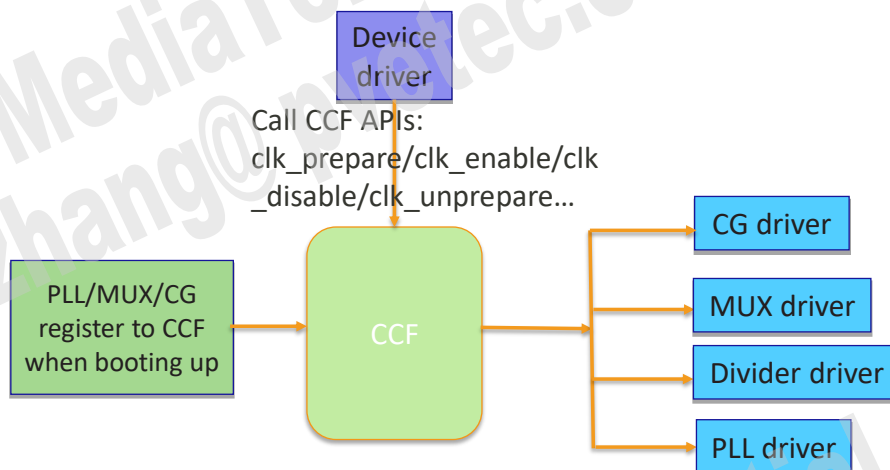


Figure 1-2. Yocto/Android/Tbox architecture

1.2.2.2 Hypervisor (Yocto + Android + Tbox)

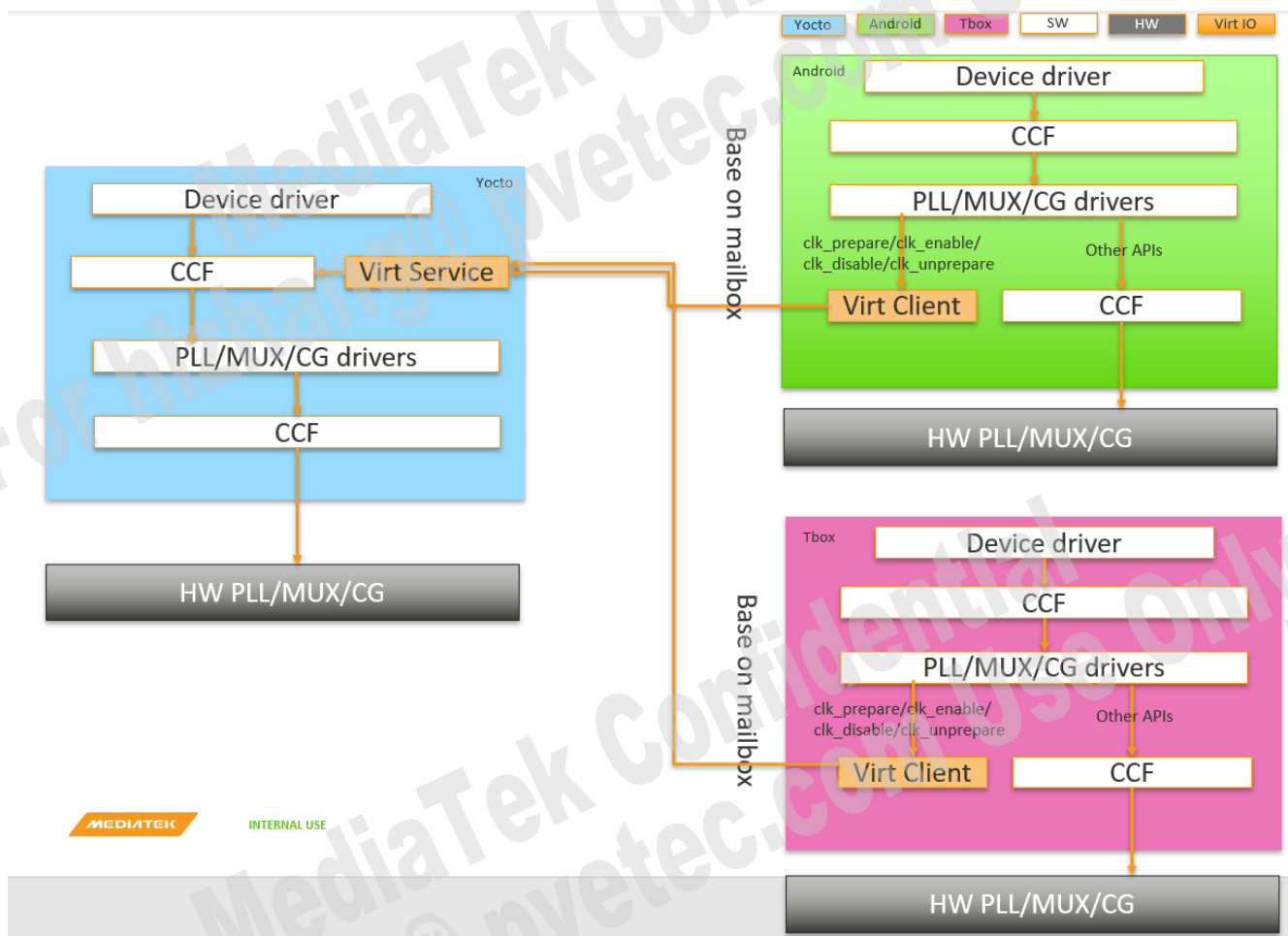


Figure 1-3. Hypervisor (Yocto + Android + Tbox) architecture

1.3 Frequently Asked Questions/Troubleshooting

1.3.1 Command to Dump Clock

```
clkdbg() { echo $@ > /proc/clkdbg ; cat /proc/clkdbg ; }
clkdbg dump_clks | grep xxxxx
```

The format of output is as follows:

The 1st column represents the clock name, the 2nd column represents On/off, the 3rd column represents is_prepared, the 4th column represents is_enabled, the 5th column represents the clock frequency, and the last column represents the parent name of current clock.

[topckgen: mdp1_ck	:	ON,	0,	0,	687499969,	mdp1_sel]
[dispsys0_config: dispsys0_mdp_aa10	:	ON,	1,	1,	687499969,	disp0_ck]
[dispsys0_config: dispsys0_mdp_rdma0	:	ON,	0,	0,	687499969,	disp0_ck]

1.3.2 Command to Query Clock Frequency

```
clkdbg() { echo $@ > /proc/clkdbg ; cat /proc/clkdbg ; }
clkdbg fmeter | grep xxxxx
```

The output is as follows:

The 1st column represents the clock name, and the 2nd column represents the frequency value, 0 means the clock is turned off.

```
fm_mdp0_ck          : 0
fm_mdp1_ck          : 0
fm_mdpl11_fs26m_guide : 26000
```

1.3.3 Command to Enable/Disable Clock

Command to enable clock:

```
clkdbg() { echo $@ > /proc/clkdbg ; cat /proc/clkdbg ; }
clkdbg prepare_enable xxxxx
```

e.g.

```
clkdbg prepare_enable mdp0_sel
clk_prepare_enable(mdp0_sel) : 0
```

The 1st line is command, and the 2nd line is output: 0 means returning OK, otherwise returning error code.

Upon dumping again, it can be seen that the above command has turned on the corresponding clock.

```
root@auto8676p164:~# clkdbg dump_clks | grep mdp0_sel
[ topckgen: mdp0_sel          : ON, 1, 1, 687499969, mmp11_d4]
```

Command to disable clock:

```
clkdbg() { echo $@ > /proc/clkdbg ; cat /proc/clkdbg ; }
clkdbg disable_unprepare xxxxx
```

1.3.4 Debug Method for Clock Hanging when Accessing HW Registers

The most common issue of the clock is hanging when accessing a related HW register, this is usually caused by the power of the current clock or the power domain not being turned on.

- If the parent clock is not turned on, the above command can be used to dump the clock to confirm whether the parent clock is normally turned on, or whether it has been turned off by other programs.
- If the power domain of clock is not turned on, the command of dumping power domain can be used to check the power domain status.

```
Cat clkdbg() { echo $@ > /proc/clkdbg ; cat /proc/clkdbg ; }
clkdbg dump_genpd | grep xxxxx
```

e.g.

```
Cat clkdbg() { echo $@ > /proc/clkdbg ; cat /proc/clkdbg ; }
clkdbg dump_genpd | grep adsp
```

The format of result is as follows:

The 1st column is power domain name, and the 2nd column is the status: active means on, suspended and POWER_OFF mean off.

The sub-nodes of power domain are names and status of sub-devices.

```
root@auto8676p164:~# clkdbg dump_genpd
+ [ovl1-shutdown          ACTIVE]
  - (power-domain-chk-41   0,   suspended)
  - (1460e000.smi-ovl1-sram-sub-comm1
0,   suspended)
- [csi-rx                  POWER_OFF]
  - (power-domain-chk-42   0,   suspended)
  - (disable-unused:disable-unused-pd-csi-rx@0
0,   suspended)
  - (genpd:1:1a00e000.seninf-top
0,   suspended)
```

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